Two-Stream Convolutional Networks for Hyperspectral Target Detection

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***Abstract*— In this article, a two-stream convolutional network-based target detector (denoted as TSCNTD) for hyperspectral images is proposed. The TSCNTD utilizes the two-stream convolutional networks to extract abundant spectral information in hyperspectral images. For the background samples, the TSCNTD finds enough typical background pixels via a hybrid sparse representation and classification-based pixel selection strategy in the entire image. To tackle the problem under limited target samples, a novel synthesis method is proposed to generate sufficient target samples with a target priori and some typical background pixels. Once the target and background samples are obtained, then the designed two-stream convolutional networks were trained with a target priori, target samples, and background samples. During training, a target priori and a target sample, which construct a positive training sample, are considered as two inputs of the two-stream convolutional networks, while a target priori and a background sample construct a negative training sample. During testing, the test samples, which are constructed by a target priori and the detected pixels, are classified by the well-trained network. The outputs of the network constitute the final detection result of the TSCNTD. Extensive experiments were made on four benchmark hyperspectral images. The experimental results indicate that the TSCNTD can achieve superior performances in target detection.**

***Index Terms*— Convolutional neural networks, deep learning, hyperspectral imagery (HSI), target detection, two-stream networks.**

I. Introduction

HYPERSPECTRAL imagery (HSI) contains extensive spectral bands and presents a high spectral resolution, so it is capable of furnishing rich spectral information [1]–[4]. Compared with traditional multispectral images, this property makes it identify different materials more accurately [5], [6]. Therefore, hyperspectral images develop numerous applications in various fields, i.e., mineral identification, urban planning, change detection, and land cover classification [7]–[10]. In these applications, detecting interesting targets has drawn increasing attention, and target detection becomes one of the most significant applications in HSI [11]–[14].

Hyperspectral target detection is a method to detect and locate targets in the image under the condition of the given target priori [15], [16]. Over the past few decades, vast detectors have emerged for target detection in HSI. In [17], a simple but effective method called spectral angle mapper (SAM), whose output is evaluated by the angle in spectral dimension between the target priori and detected pixels, is presented. According to the linear mixing model, adaptive coherence estimator (ACE) [18], [19], orthogonal subspace projection (OSP) [20], [21], and matched subspace detector (MSD) [22] are successfully applied for target detection in HSI. In [23], a well-known method named spectral matched filter (SMF) is proposed. The SMF utilizes the spectral signature for matching and achieves a good detection performance. In [24], a method with the same format of the SMF is conducted via filtering techniques. The method is constrained energy minimization (CEM). These classical detectors are usually simple and can detect most of the target pixels in hyperspectral images.

Over the years, many methods blossomed in hyperspectral target detection. In [25]–[27], the kernel constrained energy minimization (KCEM), the kernel spectral matched filter (KSMF), and the kernel target-constrained interference-minimized filter (KTCIMF) are proposed as target detectors based on kernel technique. In [28], a well-known algorithm named the sparsity-based target detector (STD) is presented based on sparsity representation. Li *et al.* [29] proposed a method combined sparse and collaborative representation (CSCR) for target detection in HSI. Following the sparsity representation and collaborative representation techniques, many scholars studied in hyperspectral target detection and developed numerous novel methods. For the purpose of achieving a purer background dictionary, Du *et al.* [30] presented a hybrid sparsity and statistics-based method (HSSD), and Lu *et al.* [16] proposed a hybrid sparsity and distance-based method (HSDD). For the sake of acquiring an optimal target priori and solving limited target pixels, *a priori* target signature optimization-based target detector (PTSOTD) is proposed in [31], and target dictionary construction-based

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sparsity detectors are proposed in [32]. To utilize the spatial information of HSI for target detection, Zhang *et al.* [33] proposed a spatially adaptive sparse representation method (SASTD), and Imani [34] provided the sparse and collaborative representation-based attribute profile method. To achieve a more discriminative residual function, Zhang *et al.* [35] presented a new sparse representation-based binary hypothesis (SRBBH) method, Bitar *et al.* [36] proposed a sparse representation and low-rank matrix decomposition-based method, and Zhu *et al.* [37] provided a binary-class collaborative representation for target detection (BCRD). Also, there are many algorithms that explore or improve from other different perspectives. Li *et al.* [38] proposed an unsupervised background extraction-based target detection method that is implemented on field-programmable gate array (FPGA) and can efficiently and robustly extract targets from hyperspectral images in real time. Dong *et al.* [39] proposed a maximum margin metric learning-based target detector (MMML), which develops a proper metric for measuring the separability between targets and backgrounds. These advanced algorithms usually perform better than the initial classical detectors.

In recent years, some scholars have paid attention to the hyperspectral target detection task using deep learning-based methods [40]. However, owing to the reason that there are not enough target samples and background samples for training deep network in hyperspectral target detection tasks, efficient works are rare. Du *et al.* [41] introduced a convolutional neural network-based target detector (CNNTD). The CNNTD utilizes the spectral difference between pixel-pairs as the input of the model and tries to distinguish between targets and backgrounds using the designed deep network. However, there are several major problems in the CNNTD. First, the CNNTD does not solve the problem if there are not enough target samples and background samples for training a deep network, and it tends to regard the target detection task as the classification task, which is not reasonable. Second, it uses the spectral difference between pixel-pairs as the input of the deep network. Compared with the original spectral signature of pixel-pairs, the spectral difference between pixel-pairs contains less discriminative spectral feature information because of the subtraction operation, which makes it harder to distinguish between targets and backgrounds and achieves a bad detection performance. Another point is that the CNNTD needs to test the spectral difference of numerous pixels to get the result of a detected pixel, which costs huge time for the detection procedure. Zhang *et al.* [42] proposed a novel deep convolutional neural network-based method for hyperspectral target detection. In [42], the method is denoted as HTD-Net. In the HTD-Net, a modified autoencoder that borrows the idea from U-Net [43] is presented to generate target samples, and a linear prediction strategy is utilized to select the background samples for the image. Also, the spectral differences between pixel-pairs constructed by these target and background pixels are fed into the designed convolutional neural network to learn the discriminative ability. Compared with the CNNTD, the logic structure of the HTD-Net is more reasonable. However, there are still some problems with the HTD-Net. The first point, as with the CNNTD, is that the HTD-Net also utilizes the spectral difference between pixel-pairs as the input of the deep network, which decreases the discriminative spectral feature between targets and backgrounds and results in poor detection performance. Besides, the HTD-Net does not simulate the subtle spectral differences between target pixels by the modified autoencoder, and the difference between generated target samples is caused by the incompleteness of the network. Moreover, the selected background samples are not complete by the linear prediction strategy, and these background samples do not include each background feature category in the image, which reduces the ability to distinguish some background pixels in the image.

To overcome the aforementioned problems and utilize the deep network to distinguish subtle spectral differences between targets and backgrounds, this article proposed a two-stream convolutional network-based target detector (denoted as TSCNTD) for hyperspectral image. In order to generate enough target samples that are sufficient to train a deep network, we proposed a novel synthesis method using the target priori and typical background pixels. In this way, it not only tackles the limited target samples but also simulates the subtle spectral differences between target pixels in hyperspectral images, which enhances the performance of highlighting target pixels. To ensure the selected background samples are more complete, the background samples are selected from the whole HSI via a proposed hybrid sparse representation and classification-based pixel selection strategy in TSCNTD. Once sufficient target samples and background samples are obtained, we can construct training samples using the target priori and these obtained samples. Then the training samples are fed into the well-designed two-stream convolutional networks to learn the discriminative ability. The two-stream convolutional networks utilize the pixel-pairs with original spectral signature as the training samples for network training, which avoids using the spectral difference between pixel-pairs. When designing the architecture of the network, we proposed an intercategory separation (ICS) loss. The ICS loss can contribute to separating targets from backgrounds so as to enhance the detection performance. After training, a well-trained network is obtained and we can get the final detection result through it. For a detected pixel, a test sample is constructed by a target priori and a detected pixel. Then the test sample is classified by the well-trained two-stream convolutional networks, and the final detection result of the detected pixel is the output of the network. The following are the summarized contributions of this article.

1) A two-stream convolutional neural network-based detector is proposed for hyperspectral target detection. The TCSNTD utilizes the well-designed two-stream convolutional neural networks to extract rich spectral information of HSI and discrimination between targets and backgrounds. Instead of using spectral difference between pixel-pairs, the pixel-pairs with original spectral signature are directly used for feature extraction. In this way, the discriminative spectral feature information of the pixel can be retained to the greatest extent, which makes it easier to distinguish between targets and backgrounds.

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2) A hybrid sparse representation and classification-based pixel selection strategy is proposed for background sample selection. Compared with other background sample selection methods, the background samples selected by the proposed strategy are more complete and pure.
3) A novel synthesis approach is presented to tackle the limited target samples. The synthesis approach generates target samples by combining the target priori with a fraction of the typical background samples. In this way, it not only tackles the limited target samples but also simulates the subtle spectral differences between target pixels in hyperspectral images.
4) In order to improve the effect of separability between targets and backgrounds, an ICS loss is proposed. The ICS loss helps separate targets from backgrounds and finally improves the detection performance.

The following is the remainder of this article. In Section II, we first review some previous algorithms. Then we detail the TSCNTD in Section III. Extensive experiments and analyses are provided in Section IV to assess the validity of the TSCNTD. Finally, some concluding remarks are presented in Section V.

II. Sparsity-Based Target Detector

The classical sparsity-based detector (STD) [28] is proposed on the basis of sparse representation. It supposes a sample of HSI could be approximated via very few elements in the complete dictionary. Consider the given hyperspectral cube by $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^{M}$ in R^d, where d presents the number of spectral bands, M expresses the number of pixels, and $\mathbf{x}_i$ represents the pixel in HSI. If the test sample $\mathbf{x}$ is a target pixel, its spectrum will lie in the target samples' composed subspace, and the test sample can then be approximated as follows:

$$\begin{aligned}\mathbf{x} &\approx \alpha_1^t\mathbf{a}_1^t + \alpha_2^t\mathbf{a}_2^t + \cdots + \alpha_{N_t}^t\mathbf{a}_{N_t}^t \\ &= \begin{bmatrix}\mathbf{a}_1^t & \mathbf{a}_2^t & \cdots & \mathbf{a}_{N_t}^t\end{bmatrix}\begin{bmatrix}\alpha_1^t & \alpha_2^t & \cdots & \alpha_{N_t}^t\end{bmatrix}^T \\ &= \mathbf{A}^t\boldsymbol{\alpha}^t \end{aligned} \tag{1}$$

where $\mathbf{A}^t$ refers to the target dictionary, $\boldsymbol{\alpha}^t$ expresses the abundances of the training samples in the $\mathbf{A}^t$, and N_t denotes the number of the target samples. In the sparse representation model, $\mathbf{A}^t$ is composed of target samples. Similarly, a background pixel would lie in the background samples' composed subspace. The test sample can then be approximated as follows:

$$\begin{aligned}\mathbf{x} &\approx \alpha_1^b\mathbf{a}_1^b + \alpha_2^b\mathbf{a}_2^b + \cdots + \alpha_{N_b}^b\mathbf{a}_{N_b}^b \\ &= \begin{bmatrix}\mathbf{a}_1^b & \mathbf{a}_2^b & \cdots \mathbf{a}_{N_b}^b\end{bmatrix}\begin{bmatrix}\alpha_1^b & \alpha_2^b & \cdots & \alpha_{N_b}^b\end{bmatrix}^T \\ &= \mathbf{A}^b\boldsymbol{\alpha}^b \end{aligned} \tag{2}$$

where $\mathbf{A}^b$ denotes the background dictionary, $\boldsymbol{\alpha}^b$ refers to the abundances of the training samples in the $\mathbf{A}^b$, and N_b is the number of samples in $\mathbf{A}^b$. Therefore, $\mathbf{x}$ would be modeled to consist in the union dictionary ((composed of $\mathbf{A}^t$ and $\mathbf{A}^b$)-composed subspaces. It can be approximately represented as

$$\begin{aligned}\mathbf{x} &\approx \left(\alpha_1^t\mathbf{a}_1^t + \alpha_2^t\mathbf{a}_2^t + \cdots + \alpha_{N_t}^t\mathbf{a}_{N_t}^t\right) \\ &\quad + \left(\alpha_1^b\mathbf{a}_1^b + \alpha_2^b\mathbf{a}_2^b + \cdots + \alpha_{N_b}^b\mathbf{a}_{N_b}^b\right) \\ &= \begin{bmatrix}\mathbf{a}_1^t & \mathbf{a}_2^t & \cdots & \mathbf{a}_{N_t}^t\end{bmatrix}\begin{bmatrix}\alpha_1^t & \alpha_2^t & \cdots & \alpha_{N_t}^t\end{bmatrix}^T \\ &\quad + \begin{bmatrix}\mathbf{a}_1^b & \mathbf{a}_2^b & \cdots & \mathbf{a}_{N_b}^b\end{bmatrix}\begin{bmatrix}\alpha_1^b & \alpha_2^b & \cdots & \alpha_{N_b}^b\end{bmatrix}^T \\ &= \mathbf{A}^t\boldsymbol{\alpha}^t + \mathbf{A}^b\boldsymbol{\alpha}^b = \mathbf{A}\boldsymbol{\alpha} \end{aligned} \tag{3}$$

where $\mathbf{A}$ is a union dictionary, and $\boldsymbol{\alpha}$ is the concatenation of $\boldsymbol{\alpha}^t$ and $\boldsymbol{\alpha}^b$. The aforementioned sparse vector $\boldsymbol{\alpha}$ is acquired by settling the following minimization equation:

$$\widehat{\boldsymbol{\alpha}} = \operatorname{argmin} \|\mathbf{A}\boldsymbol{\alpha} - \mathbf{x}\|_2 \quad \text{s.t.} \left\|\widehat{\boldsymbol{\alpha}}\right\|_0 \le K \tag{4}$$

where $||\cdot||_0$ expresses ℓ_0-norm, it can be defined as the number of nonzero entries in $\widehat{\boldsymbol{\alpha}}$ [28], and K refers to a preset upper limit value in the sparsity level. The equation in (4) has been proved to be a NP-hard problem. Therefore, the orthogonal matching pursuit (OMP) [44], [45] method is utilized to tackle (4) in this article.

After solving the NP-hard problem, the sparse vector is obtained. Then, the class label of the test sample depends on reconstructing the features of the sparse vector. The process of reconstruction can be expressed as follows. First, the vector $\widehat{\boldsymbol{\alpha}}$ is decomposed into $\widehat{\boldsymbol{\alpha}}^t$ and $\widehat{\boldsymbol{\alpha}}^b$. Then according to $\widehat{\boldsymbol{\alpha}}^t$ and $\widehat{\boldsymbol{\alpha}}^b$, the residual of recovery can be acquired by

$$\begin{aligned} r_t(\mathbf{x}) &= \left\|\mathbf{x} - \mathbf{A}^t\widehat{\boldsymbol{\alpha}}^t\right\| \\ r_b(\mathbf{x}) &= \left\|\mathbf{x} - \mathbf{A}^b\widehat{\boldsymbol{\alpha}}^b\right\|. \end{aligned} \tag{5}$$

Algorithm 1 TSCNTD

Input: The hyperspectral image and the target priori.

1. Select typical background samples via the hybrid sparse representation and classification-based pixel selection strategy.
2. Utilize background samples and target priori to generate target samples.
3. Construct training samples: positive training samples with label 1 that are constructed by target priori and target samples, negative training samples with label 0 that are constructed by target priori and background samples.
4. Train the well designed two-stream convolutional neural networks using the training samples.
5. Construct test samples by combining target priori and detected pixels.
6. Feed the test samples into the well trained network.
7. Obtain the output of the network for each detected pixel and constitute the final detection result.

Output: The detection results of the proposed TSCNTD.

Finally, the final output of the STD method is obtained by

$$D_{\text{STD}}(\mathbf{x}) = r_b(\mathbf{x}) - r_t(\mathbf{x}) \tag{6}$$

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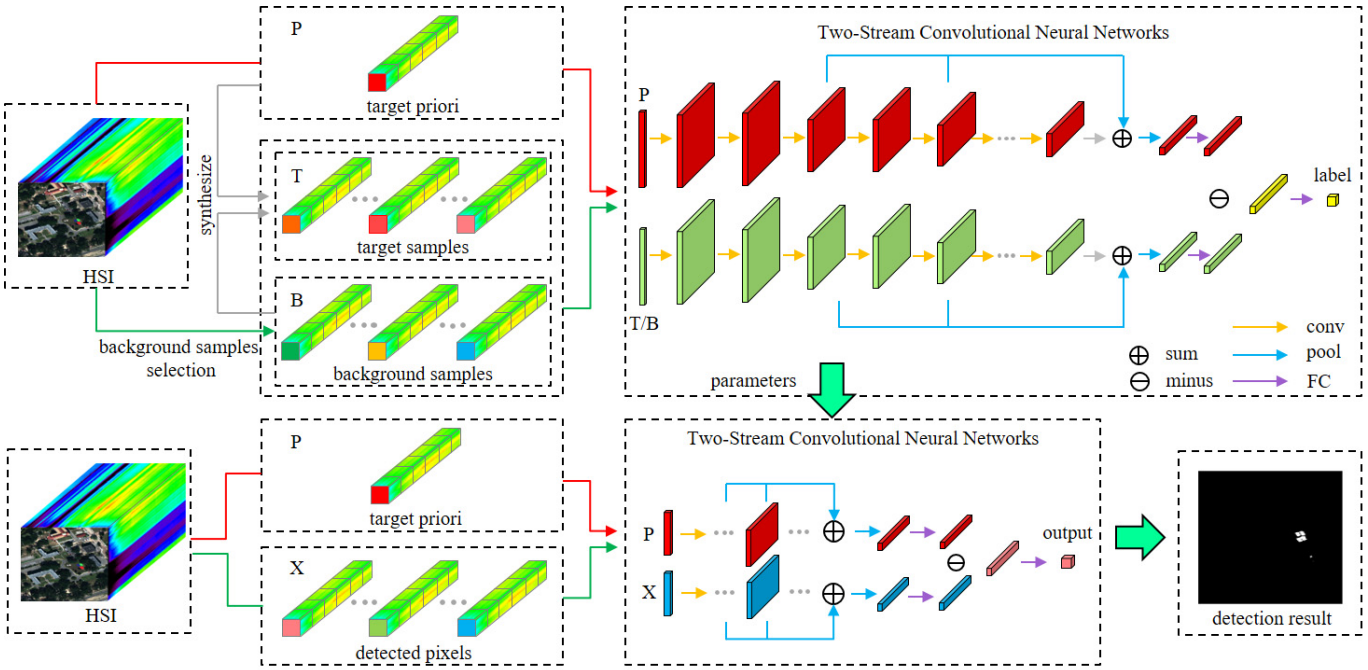

Fig. 1. Flowchart of the proposed TSCNTD. Before training, a hybrid sparse representation and classification-based pixel selection strategy is proposed to select the typical background samples in HSI, then sufficient target samples are generated by some typical background samples and target priori. During training, the training samples (positive training samples with label 1 that are constructed by the target priori and target samples, and negative training samples with label 0 that are constructed by the target priori and background samples) are fed into the well-designed two-stream convolutional networks to learn the discriminative ability. During testing, the test samples (constructed by the target priori and detected pixels) are classified by the well-trained two-stream convolutional networks. The outputs of the network constitute the final detection results.

III. Two-Stream Convolutional Networks for Target Detection

In this section, the two-stream convolutional neural network-based target detector (TSCNTD) is described in detail. Fig. 1 illustrates the flowchart of the TSCNTD. The whole flowchart can be divided into two stages: the training stage and the testing stage. The training stage includes background sample selection, target sample generation, training sample construction, and training the two-stream convolutional neural networks. In the testing stage, detected pixels are combined with the target priori to construct the test samples. These test samples are classified by the well-trained network. Then the proposed TSCNTD obtains the final detection results. The detailed implementation of the proposed TSCNTD is presented as Algorithm 1.

A. Background Sample Selection

To accurately estimate the background information of HSI, we proposed a hybrid sparse representation and classification-based pixel selection strategy for background sample selection. Fig. 2 presents the flowchart of the pixel selection strategy.

For a given HSI, the principal component analysis (PCA) is applied to the imagery. By the reason that some pixels in different local regions show the same spectrum, which shows nonlocal similarity among local regions [11], the nonlocal similarity is considered when selecting background samples. Therefore, the pixels of the HSI are grouped into m clusters through the K-means algorithm. In the ith cluster, there are m_i pixels which can be written as

$$\mathbf{E}_i = \{\mathbf{e}_{i1}, \mathbf{e}_{i2}, \ldots, \mathbf{e}_{im_i}\}, \quad 1 \le i \le m. \tag{7}$$

For each cluster, we want to pick out those pixels that have typical background features. Inspired by the sparse representation model, each pixel $\mathbf{e}_{is}$ in the ith cluster, where $1 \le s \le m_i$, is approximately represented as a linear combination of the pixels $\bar{\mathbf{E}}_i$(except pixel $\mathbf{e}_{is}$) in this cluster which can be expressed by

$$\begin{aligned} &\mathbf{e}_{is} = \bar{\mathbf{E}}_i \theta_{is}, \quad 1 \le i \le m, 1 \le s \le m_i \\ &\text{s.t. } \bar{\mathbf{E}}_i = \{\mathbf{e}_{i1}, \mathbf{e}_{i2}, \ldots, \mathbf{e}_{i(s-1)}, \mathbf{e}_{i(s+1)}, \ldots, \mathbf{e}_{im_i}\} \end{aligned} \tag{8}$$

where $\boldsymbol{\theta}_{is}$ denotes the abundances of samples in $\bar{\mathbf{E}}_i$ and it is a sparse vector. The method to recover it can refer to the STD in Section II. For the ith cluster, the number of sparse vectors we can obtain is m_i. Then, we simply calculate the frequency score of $\boldsymbol{\theta}_{is}$ using ℓ_1-norm. The corresponding $\mathbf{e}_{is}$ is a more typical background pixel in the ith cluster if $\boldsymbol{\theta}_{is}$ has a higher frequency score [31]. We pick out these typical background pixels as background samples in this cluster. For each cluster, a certain percentage of background pixels are selected. All typical background pixels selected from each cluster finally constitute the whole background samples of the given HSI. Here, we use N that represents the total number of background samples and its formula is

$$N = \sum_{i=1}^{m} j_i \tag{9}$$

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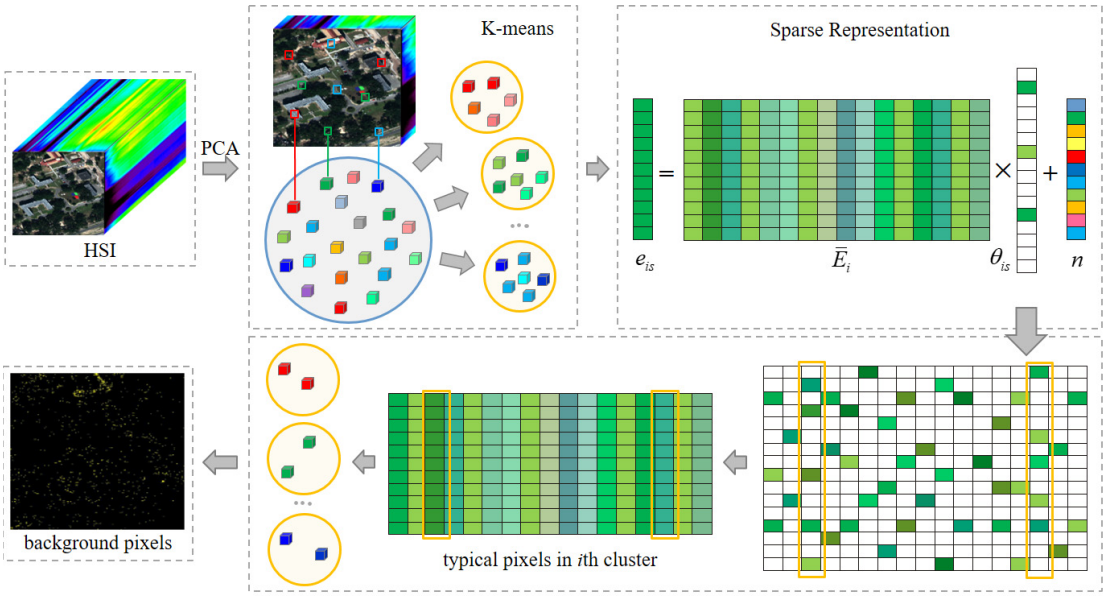

Fig. 2. Flowchart of background sample selection.

where j_i denotes the number of the selected pixels in the ith cluster. Once N is determined, the percentages in each cluster can be calculated according to the pixels of the given HSI in all.

Taking advantage of the nonlocal similarity among the local regions, background samples selected in the given HSI are more complete, which means that there include almost all kinds of background features in the background samples. Moreover, for the reason that target pixels account for only a small part and background pixels account for a large part, the target pixel cannot be a typical pixel when selecting pixels in each cluster. Therefore, background samples selected via the hybrid sparse representation and classification-based pixel selection strategy are not only complete but also pure, which enhances the final detection performance.

B. Target Sample Generation

In order to tackle the problem that there are limited target samples which are not enough to meet the need of training deep network in HSI target detection task, a new synthesis method is proposed to generate sufficient target samples. Due to the limited spatial resolution, subpixels, where more than one material appears in a pixel, are widespread in the hyperspectral image [46]. In the HSI target detection task, subpixel targets are also regarded as target pixels to be detected. Therefore, we utilize the target priori and typical background samples obtained from background sample selection to synthesize subpixel targets. Specifically, for a target priori pixel, a certain abundance of the background material (typical background samples) is implanted in the target priori pixel. The abundance of background material is randomly set as 0%–20% in a subpixel. Considering the fact that there exist pure target pixels in HSI, some target priori pixels contain no background material (when the abundance is set as 0%). These pixels are regarded as pure target samples. As a result, for a target priori pixel, N target samples can be generated via the synthesis method since the total number of typical background samples is N. Based on the synthesis method, sufficient target samples can be generated for training deep networks. Moreover, it simulates the subtle spectral differences between target pixels in hyperspectral images, and more subpixel targets in HSI can be detected since we have fully considered the case of subpixels in our proposed synthesis method.

C. Training Sample Construction

Based on the aforementioned background sample selection and target sample generation, we can obtain enough typical background samples and target samples. Before training a deep network, we construct training samples using these obtained samples and target priori. Two types of training samples are constructed. They are positive training samples with label 1 that are constructed by the target priori and target samples, and negative training samples with label 0 that are constructed by the target priori and background samples. Specifically, a target priori pixel and a target sample construct a positive training sample, while a negative training sample is constructed by a target priori pixel and a background sample.

Hensman and Masko [47] show that the performance of CNN is influenced by the distribution of training samples. As suggested in this article, the balanced distribution of the training samples yields the best performance. In our work, for a target priori pixel, N target samples can be generated by N background samples. Then, N target samples and the target priori pixel construct N positive training samples, while N background samples and the target priori pixel construct N negative training samples. As a result, the positive training samples' number equals the negative training samples' number, which means a balanced distribution in the training samples. Moreover, for a given HSI with p target priori pixels, the constructed negative training samples are $p * N$, and the

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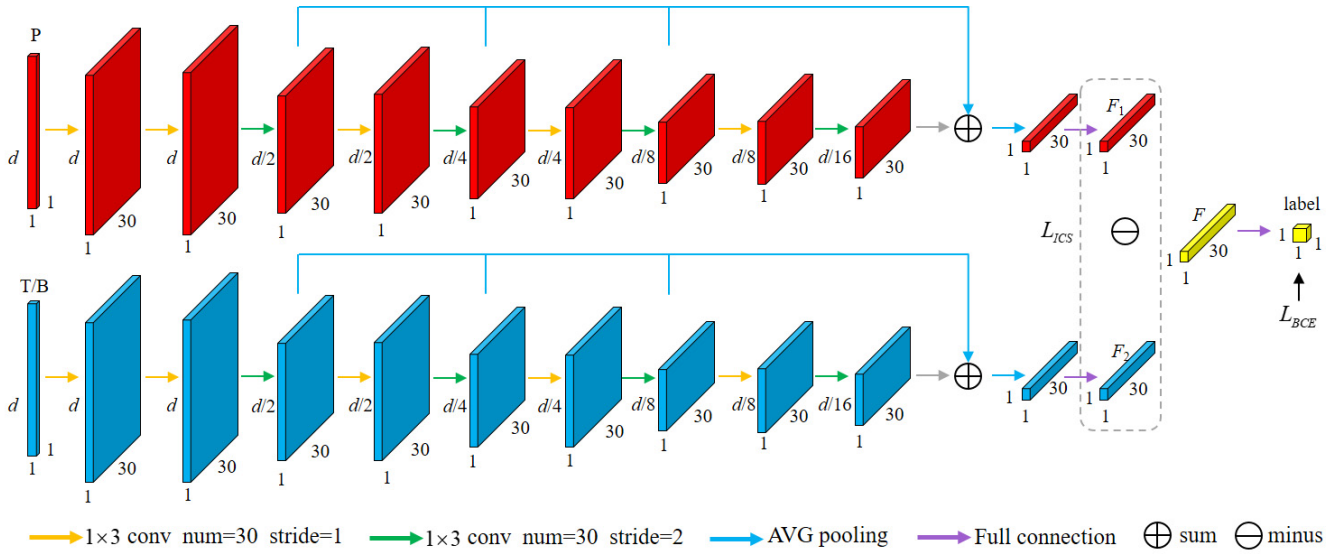

Fig. 3. Architecture and parameters of the two-stream convolutional neural networks.

total number of training samples that will be feed into the network is $2p * N$.

D. Two-Stream Convolutional Neural Networks

The architecture of the two-stream convolutional neural networks is shown in Fig. 3. The network includes two branches: up-branch and down-branch. Each branch contains an input. In each branch, nine convolutional layers are utilized to extract rich spectral information of input pixels. In this article, the 1-D convolution layer is utilized to implement the convolution operations. These convolutional layers are followed by rectified linear unit (ReLU) layers. For the reason that the pooling layer may cause spectral information loss when spectral dimension reduction is required, convolutional layers whose kernel stride equals to two are used in the network instead of using pooling layers. In order to maximize the retention of spectral features, based on different average (AVG) pooling layers, we add up all features extracted by the convolutional layers whose kernel stride is two to the feature extracted by the final convolutional layer. Then the final features of each branch are obtained by an AVG pooling layer and a full connection (FC) operation. Suppose d as the number of the spectral band, the parameters of the two-stream convolutional neural networks are shown in Fig. 3. In the network, P refers to a target priori pixel, T represents the target pixel, and B is the background pixel. According to the aforementioned training sample construction, the input of up-branch is always P, and the label of the training sample is 1 when the input of down-branch is T, otherwise, the label is 0. After some convolution operations, several pooling operations, and an FC operation, the final features of two branches are obtained (denoted as F_1 and F_2). Then we combine two features by

$$F = F_1 - F_2. \tag{10}$$

Finally, the output of the network is obtained via the last FC layer followed by a sigmoid function.

The loss functions of the network are illustrated here. There are two loss functions in the proposed network. The first one is binary cross-entropy (BCE) loss

$$L_{\text{BCE}} = -\frac{1}{n}\sum_{i=1}^{n}\left[l_i * \log(f_i) + (1 - l_i) * \log(1 - f_i)\right] \tag{11}$$

where n is the batch size, l_i denotes the training sample's label, and f_i refers to the output of the sigmoid function. The other one is the ICS loss. In order to improve the effect of separability between targets and backgrounds, the ICS loss is proposed. Fig. 4 presents the illustration of the ICS loss. If the inputs of the two branches are P and T (the label is 1), they belong to the same class and the distance between them should be minimized, otherwise, they belong to different classes and the distance between them should be maximized. Therefore, the proposed ICS loss is expressed as follows:

$$L_{\text{ICS}} = \begin{cases} \dfrac{1}{n}\displaystyle\sum_{i=1}^{n}\left(1 - \frac{\vec{F}_{1i}\cdot\vec{F}_{2i}}{\left\|\vec{F}_{1i}\right\|\left\|\vec{F}_{2i}\right\|}\right), & \text{if label} = 1 \\ \dfrac{1}{n}\displaystyle\sum_{i=1}^{n}\left(\frac{\vec{F}_{1i}\cdot\vec{F}_{2i}}{\left\|\vec{F}_{1i}\right\|\left\|\vec{F}_{2i}\right\|}\right), & \text{if label} = 0 \end{cases} \tag{12}$$

where F_1 is the extracted feature of the up-branch and F_2 is the extracted feature of the down-branch. The final loss function L is the sum of ICS loss and BCE loss, which is expressed as

$$L = L_{\text{ICS}} + L_{\text{BCE}}. \tag{13}$$

E. Detected Pixels' Testing

After the training stage, a well-trained network is obtained. Detected pixels are combined with the target priori to construct the test samples that are classified by the well-trained network. Specifically, considering there are p target priori pixels in the detected HSI, for a detected pixel, p test samples can be constructed and we can get p output values of the test samples

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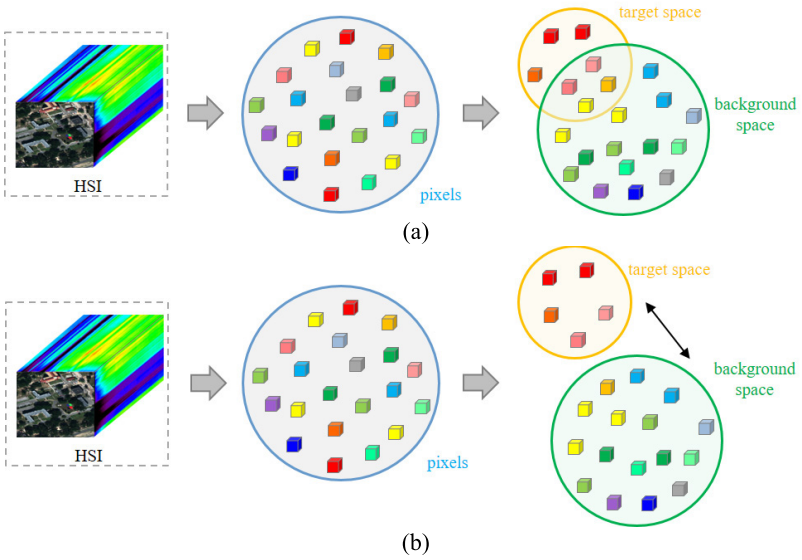

Fig. 4. Illustration of the proposed ICS loss. (a) BCE loss. (b) ICS loss and BCE loss. (a) Detection performance where only BCE loss is used in the proposed network. Most of the target pixels and background pixels can be separated, but there is still a certain overlap. (b) Detection performance using BCE loss and ICS loss. The separability performance is much better than (a). The ICS loss can contribute to separating targets from backgrounds and finally enhance the detection performance.

via the well-trained network. We simply AVG these output values and obtain the detection result of the detected pixel

$$V = \frac{1}{p}\sum_{i=1}^{p} V_i \tag{14}$$

where V_i is the output value of the test sample. The final output of TSCNTD is

$$D_{\mathrm{TSCNTD}}(\mathbf{x}) = V(\mathbf{x}). \tag{15}$$

If $D_{\mathrm{TSCNTD}}(x)$ is larger than a threshold η, x can be classified as a target pixel, otherwise, x is a background pixel.

IV. Experimental Results and Analysis

A. Hyperspectral Data Set

1) Muufl Gulfport Data Set: The first benchmark image is collected from Long Beach, MS, USA. The image scene is a part of the Gulf Park Campus at The University of Southern Mississippi, Hattiesburg, MS. In this image, the wavelength range is 367.7–1043.4 nm which covers 72 spectral bands. After discarding the noise bands (1–4 and 69–72), 64 bands remained. This image has 325 × 337 pixels in all. In this article, an image scene that covers 325 × 220 pixels is used for target detection. In this image scene, several cloth panels, which contain 269 pixels, are regarded as targets to be detected. The illustration of this data set is provided in Fig. 5(a) and (b) [48].

2) Mosaic Avon Data Set: This benchmark image is collected in the "SpecTIR Hyperspectral Airborne Experiment 2012." The image in this campaign is assembled by a push-broom sensor in the south of Rochester, NY, USA. The scene in the image is a part of the park in Avon. This image has 360 spectral bands and its wavelength range is 400–2450 nm. The data set has 1-m spatial resolution and 5-m spectral resolution. In the data set, a subscene that contains 200 × 300 pixels was chosen as experimental data to evaluate the proposed algorithm. In the subscene, there are 28 panels covering 75 target pixels, and these pixels are the detected pixels. The illustration of this data set is provided in Fig. 6(a) and (b) [49].

3) San Diego Data Set: This image is an airborne visible/infrared imaging spectrometer (AVIRIS) -assembled HSI. The scene in this image is an airport in San Diego. This image has 224 spectral bands and the resolution of this image is 3.5 m. In the San Diego data set, the wavelength range is 370–2510 nm. After discarding 35 bad bands, 189 bands are retained in the data [1]. A scene that covers 200 × 200 pixels is employed for the experiment. Fig. 7(a) presents the sence of this data set. There are 134 pixels distributed in three planes and they are regarded as targets needed to be detected which are illustrated in Fig. 7(b) [50].

4) Nuance Cri Data Set: The last image was assembled via the Nuance Cri hyperspectral sensor. The wavelength range of this data set is 650–1100 nm and there are 46 spectral bands. It covers 400 × 400 pixels as shown in Fig. 8(a). In this image, the ten rocks containing anomalous pixels different from the grassy background are illustrated in Fig. 8(b). The 1254 pixels are regarded as targets to be detected [51].

B. Experimental Details

1) Evaluation Metrics: To investigate the performance of the TSCNTD, three quantitative evaluation indexes are employed. They are receiver operating characteristic (ROC) curve, area under curve (AUC) value, and separability map.

The first evaluation index is the ROC curve [5], which is a widely used evaluation tool for assessing the performance of detectors. The ROC curve can provide a threshold-free detection performance comparison by detecting a continuous

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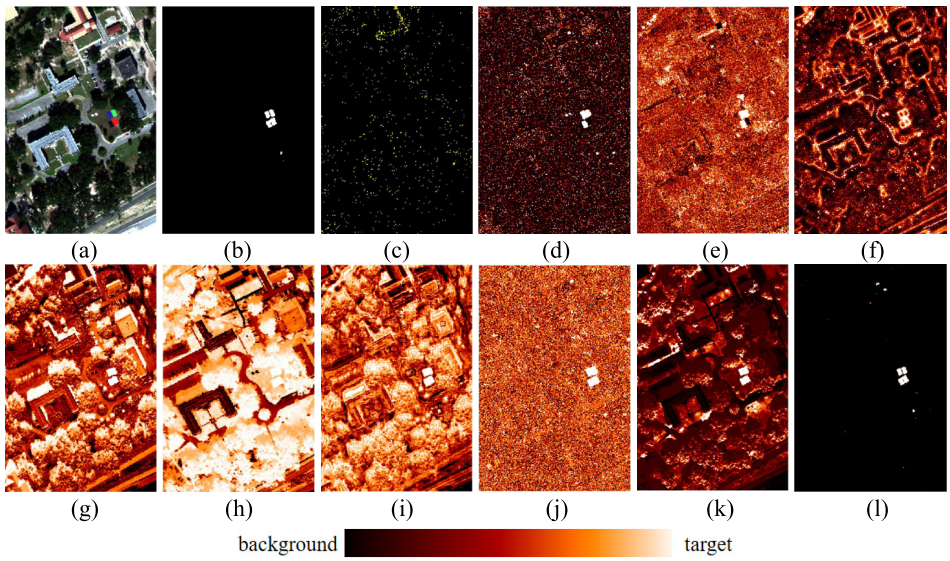

Fig. 5. Muufl Gulfport HSI and color maps of the detection results. (a) Image scene. (b) Ground truth. (c) Selected background pixels. (d) ACE. (e) CEM. (f) CNNTD. (g) STD. (h) CSCR. (i) SASTD. (j) HSSD. (k) HTD-Net. (l) TSCNTD.

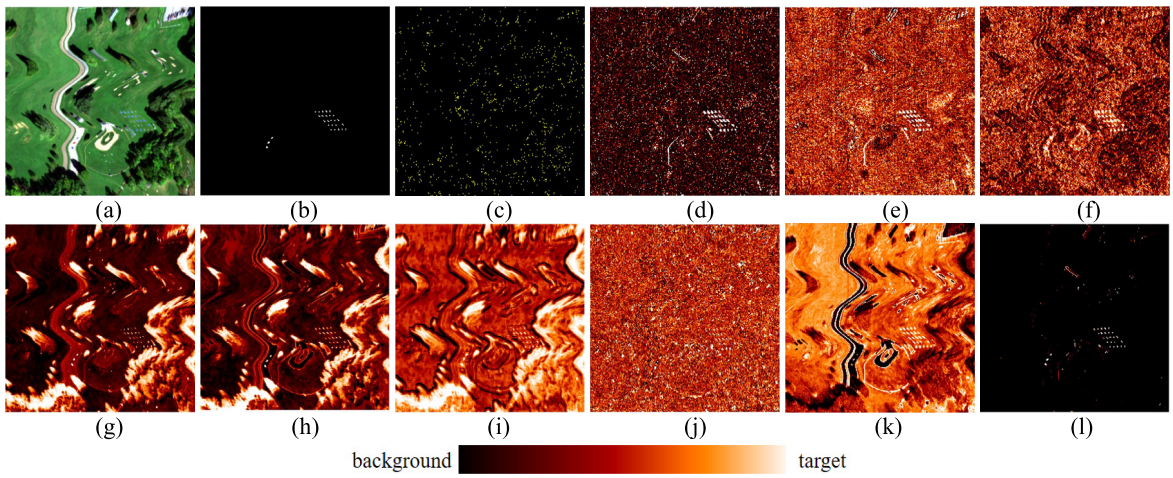

Fig. 6. Mosaic Avon HSI and color maps of the detection results. (a) Image scene. (b) Ground truth. (c) Selected background pixels. (d) ACE. (e) CEM. (f) CNNTD. (g) STD. (h) CSCR. (i) SASTD. (j) HSSD. (k) HTD-Net. (l) TSCNTD.

curve of the probability p_d and the false alarm rate p_f

$$p_d = \frac{N_{\text{detected}}}{N_t}$$
$$p_f = \frac{N_{\text{miss}}}{N_{\text{all}}} \tag{16}$$

where N_{detected} denotes the number of target pixels that have been detected, N_t refers to the number of real target pixels in the HSI, N_{miss} denotes the misjudged target pixels and background pixels, and N_{all} denotes the number of all pixels in HSI. Under the same false alarm level, the algorithms with higher detection rates have better detection performance.

The AUC value [52], [53] is an effective tool to quantitatively evaluate the detection performance. It refers to the area under the ROC curve. The AUC value can be calculated by the integral value from the ROC curve to the coordinate axis. The range of the AUC value is 0–1, and the algorithm with a higher AUC value performs better.

To assess the ability of the proposed TSCNTD in separating targets and backgrounds, this article presents another evaluation index separability map. The separability map is expressed by the normalized detection statistic range. In the separability map, pixels with detection result values are divided into two classes that are target class and background class according to the ground truth. Then, all output values of the detector are normalized to 0–1 in target and background class, respectively. A box chart is used to express the separability between two classes. In the separability map, the orange boxes present the range of background class, while the green boxes express the target class. Boxes illustrate the main portions of the pixels between 25% and 75% in their class. The algorithm with a

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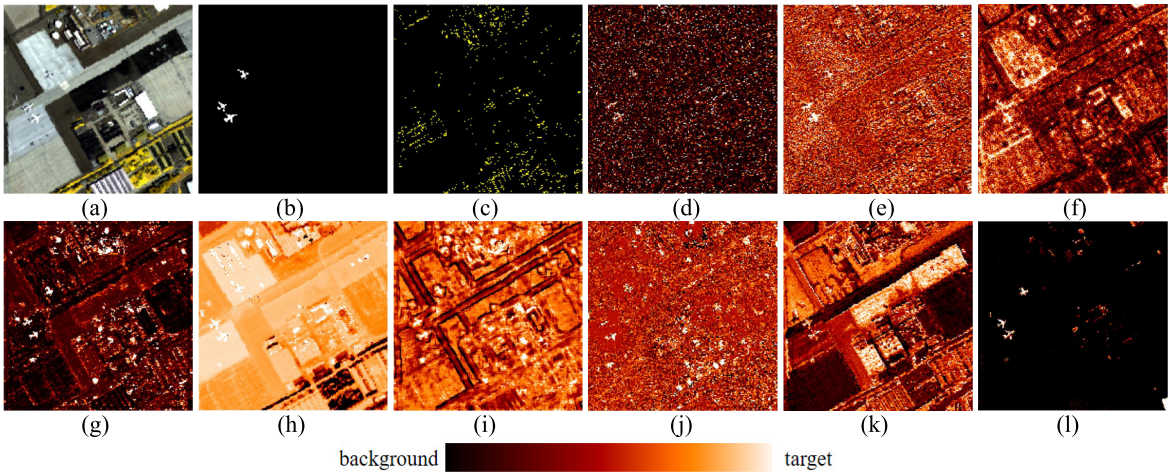

Fig. 7. San Diego HSI and color maps of the detection results. (a) Image scene. (b) Ground truth. (c) Selected background pixels. (d) ACE. (e) CEM. (f) CNNTD. (g) STD. (h) CSCR. (i) SASTD. (j) HSSD. (k) HTD-Net. (l) TSCNTD.

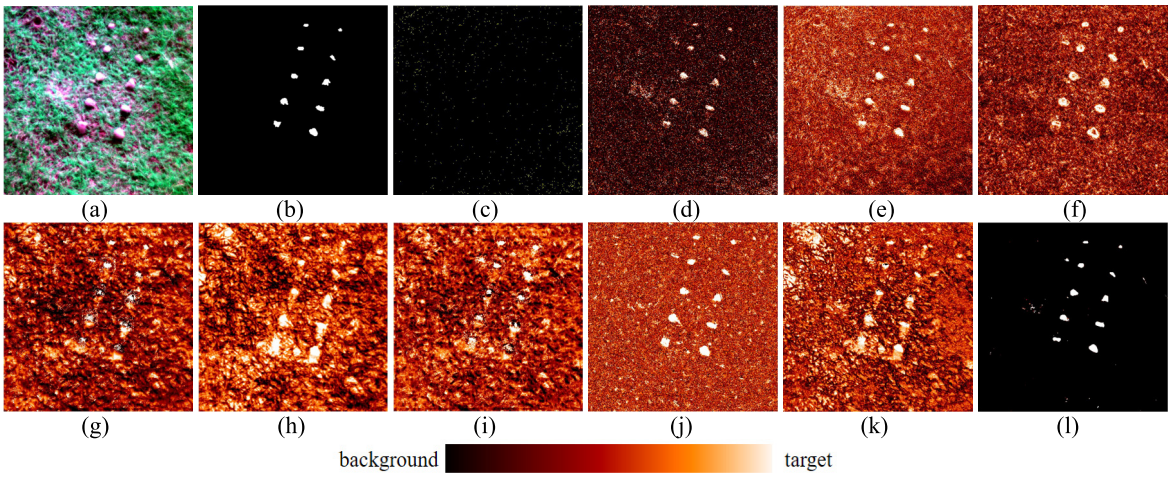

Fig. 8. Nuance Cri HSI and color maps of the detection results. (a) Image scene. (b) Ground truth. (c) Selected background pixels. (d) ACE. (e) CEM. (f) CNNTD. (g) STD. (h) CSCR. (i) SASTD. (j) HSSD. (k) HTD-Net. (l) TSCNTD.

larger separability distance between two classes and a smaller background distribution range means a better separability performance.

2) Experimental Setup: According to the description of the proposed TSCNTD, there are several critical parameters, such as the number of selected typical background samples and the abundance of background material implanted in the target priori pixels. Considering the complexity of background in HSI, the number of selected typical background samples is set as $N = 1000$ in our experiments, which can make sure background samples are complete. The (c) of Figs. 5–9 shows the selected background pixels of each image. The abundance of background material implanted in the target priori pixels is set as 0%–20%, which makes more target pixels can be detected and conduces to superior detection performance.

For the proposed two-stream convolutional neural networks, batch size and learning rate are parameters. We set the batch size as 256 to make full use of the parallel computing ability of GPU. Also, for the learning rate, when it is large, the model may not converge, and when the learning rate is too small, it could take a lot of time to converge. We set the learning rate as 1e-3 in this article, which costs relatively less time and can obtain a good training performance.

To investigate the performance of TSCNTD, eight advanced target detectors are compared with the TSCNTD. These eight target detectors are: two classical detectors, i.e., ACE [19] and CEM [24]; four advanced representation-based detectors, i.e., STD [28], CSCR [29], SASTD [33], and HSSD [30]; and two deep learning-based methods, i.e., CNNTD [41] and HTD-Net [42].

For all detectors, we utilize the same target priori for target detection. In the experiment, we achieve the parameters for the detection results in each detector by cross-validation. For the detectors that need a dual concentric window to model the local background, i.e., STD, CSCR, SASTD, and HSSD, the size of the window is various since the size of the target of interest is various from data to data. In this article, the size is set as 35×35 and 17×17 for the Muufl Gulfport

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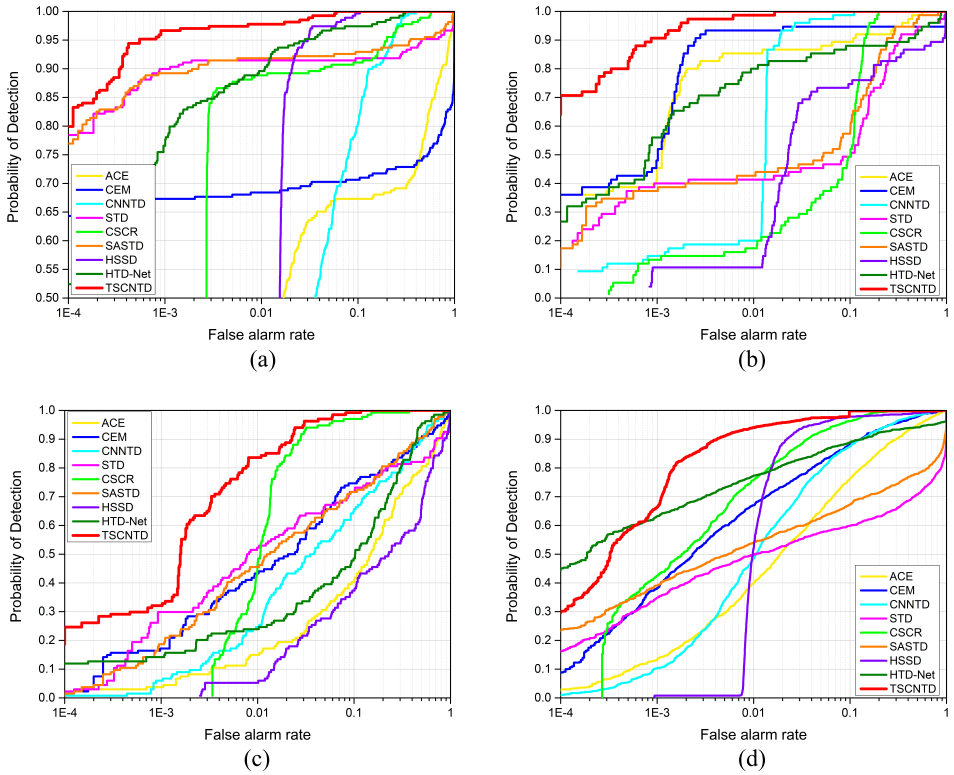

Fig. 9. ROC curves for four data sets. (a) Muufl Gulfport. (b) Mosaic Avon. (c) San Diego. (d) Nuance Cri.

data set, 15 × 15 and 5 × 5 for the Mosaic Avon data set, 21 × 21 and 11 × 11 for the San Diego data set, and 27 × 27 and 13 × 13 for the Nuance Cri data set.

C. Comparison Performance With State-of-the-Art Methods

First, the color maps of the detection results for all detectors in four HSIs are provided in (d)–(l) of Figs. 5–9. For the Muufl Gulfport image, two classical detectors ACE and CEM can detect most of target pixels, but they fail to detect the red panel. The CNNTD achieves the worst detection performance among all detectors as it just detects few target pixels. Although other detectors, i.e., STD, CSCR, SASTD, and HTD-Net, have a good performance in detecting targets, they fail to suppress the backgrounds. The HSSD performs better in suppressing backgrounds, but that is not enough. Our proposed TSCNTD can not only detect almost all targets but also perform well in suppressing backgrounds. For the Mosaic Avon data set, the representation-based methods, i.e., STD, CSCR, and SASTD, have a bad performance in suppressing backgrounds. The performance of the HSSD is poor since the HSSD cannot distinguish targets or suppress backgrounds. Two classical methods ACE and CEM even perform better, as they highlight most of targets and suppress backgrounds, but they fail to highlight three panels on the left side. Two deep learning-based methods, the CNNTD and HTD-Net, can highlight most of targets but fail to suppress some background pixels. The proposed TSCNTD method can accurately highlight all target pixels and suppress almost all backgrounds, so it achieves a fairly good detection performance. For the San Diego data set, it is evident that our TSCNTD is capable of detecting all targets with high confidence. Two classical detectors ACE and CEM and the HSSD detector fail to detect some target pixels, i.e., the airplane in the top left of the image. All other detectors fail to suppress the backgrounds and derive the detection results with more false alarms. For the Nuance Cri data set, it can be seen that several representation-based detectors and two deep learning-based methods, i.e., STD, SASTD, CNNTD, and HTD-Net, can highlight part of the target pixels. The CSCR detector is capable of identifying the targets, but the CSCR fails to suppress the boundaries in the image, which causes more false alarms. Compared to the previous methods, ACE, CEM, HSSD, and TSCNTD have significantly better performances, as they accurately extract the targets from the backgrounds. TSCNTD achieves an even more outstanding performance since the effect of background suppressing is optimal among these detectors.

Then the ROC curves and AUC values of nine detectors for four data sets are provided to quantitatively assess the detection performance of the TSCNTD, as illustrated in Fig. 9 and Table I, respectively. In Fig. 9, the red curves are the ROC curves of our proposed TSCNTD for four data sets. It is evident that TSCNTD achieves optimal ROC curves in all the four HSI data sets, as it is always ahead of the curves of other detectors, generating low false alarms and high detection probability. From Table I, we can see that the AUC values of two classical detectors ACE and CEM are not bad in Mosaic Avon and Nuance Cri data sets. Two representation-based methods, i.e., STD and SASTD, get a higher AUC value compared to ACE and CEM in most cases. However,

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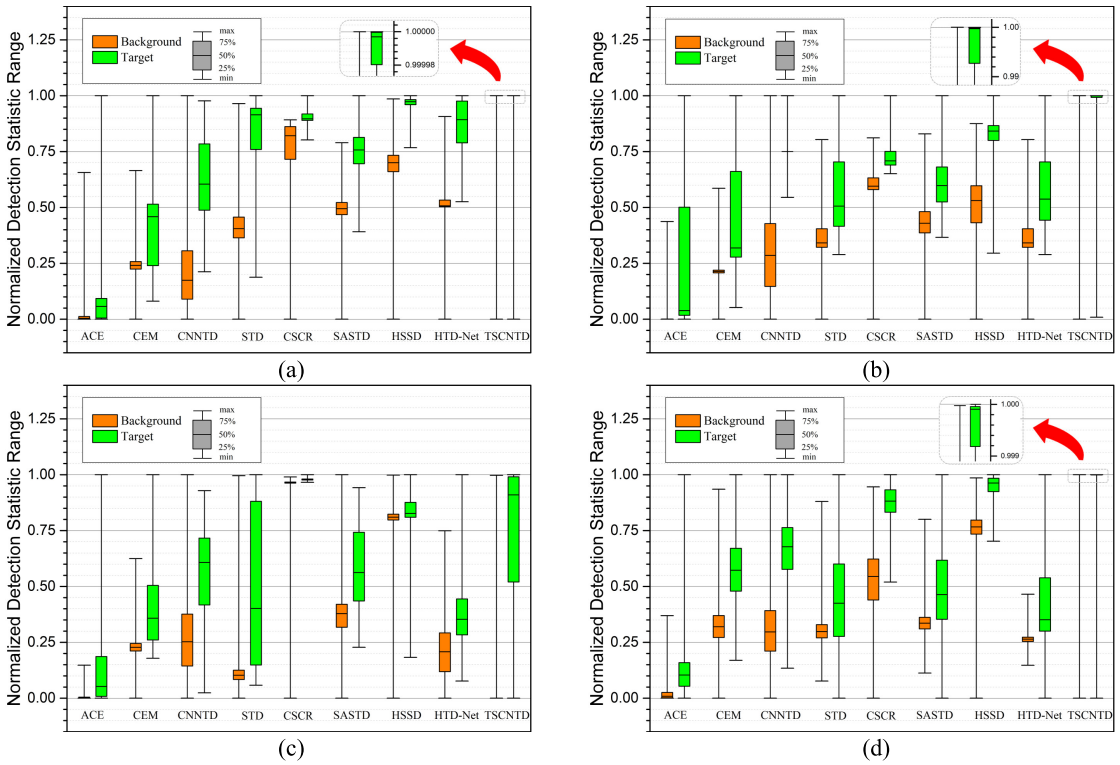

Fig. 10. Separability maps for four data sets. (a) Muufl Gulfport. (b) Mosaic Avon. (c) San Diego. (d) Nuance Cri.

TABLE I
AUC VALUES' COMPARISON FOR FOUR DATA SETS

Methods	Muufl Gulfport	Mosaic Avon	San Diego	Nuance Cri
ACE	0.8177	0.9677	0.7474	0.9059
CEM	0.7635	0.9524	0.8691	0.9549
CNNTD	0.9421	0.9871	0.8541	0.9509
STD	0.9454	0.8613	0.8356	0.7067
CSCR	0.9736	0.9203	0.9833	0.9860
SASTD	0.9562	0.9020	0.8782	0.7741
HSSD	0.9828	0.8446	0.6586	0.9777
HTD-Net	0.9923	0.9293	0.8357	0.9317
TSCNTD	**0.9991**	**0.9996**	**0.9935**	**0.9875**

the AUC values of STD and SASTD are not that high as they perform badly in suppressing backgrounds. CSCR and HSSD achieve high AUC values in Muufl Gulfport and Nuance Cri data sets since they can accurately detect targets. Also, CSCR is even better as the AUC value of CSCR for San Diego is much higher than that of HSSD. Two deep learning-based methods, the CNNTD and HTD-net perform well, especially the HTD-Net in the Muufl Gulfport data set. However, our proposed TSCNTD outperforms all other detectors in the AUC value as it yields the highest AUC values in four HSIs. Therefore, TSCNTD achieves superior detection performances in all four data sets. The superior detection performances of the TSCNTD in four HSIs show that the hybrid sparse representation and classification-based pixel selection strategy can provide complete and pure typical background sample, while the synthesis method can solve the limited target samples providing sufficient target sample for training a deep network, and the two-stream convolutional neural network can identify the targets and successfully extract the target pixels from the background pixels, therefore accurately suppress backgrounds and detecting targets.

Moreover, to test the validity of TSCNTD in separating targets and backgrounds, we provide the normalized detection statistic range for four HSI data sets via separability maps. The separability maps are presented in Fig. 10. It is obvious that for all the four HSI data sets, the TSCNTD has an outstanding performance in suppressing backgrounds as the orange boxes of them are low and they all present small background distribution ranges. The TSCNTD also achieves excellent performance in highlighting targets since the green boxes of the TSCNTD for four data sets are high, especially in Muufl Gulfport, Mosaic Avon, and Nuance Cri data sets. There exists a larger gap between two boxes of the detector TSCNTD than those of other detectors in all the four HSI data sets, as the green boxes and orange boxes of the other detectors are overlapped or present a small gap. As a result, the proposed TSCNTD yields the best performance in separating targets from backgrounds. The superior separability indicates that the proposed ICS loss can reduce the intraclass distance and increase the interclass distance, which finally contributes to separating targets from backgrounds.

D. Analysis of ICS Loss

To explain the effectiveness of the ICS loss, we compare the detection performances and separability maps of the TSCNTD with those of the TSCNTD without ICS loss. Fig. 11 shows the comparison of ROC curves with BCE loss and ICS loss for four data sets. The comparison of

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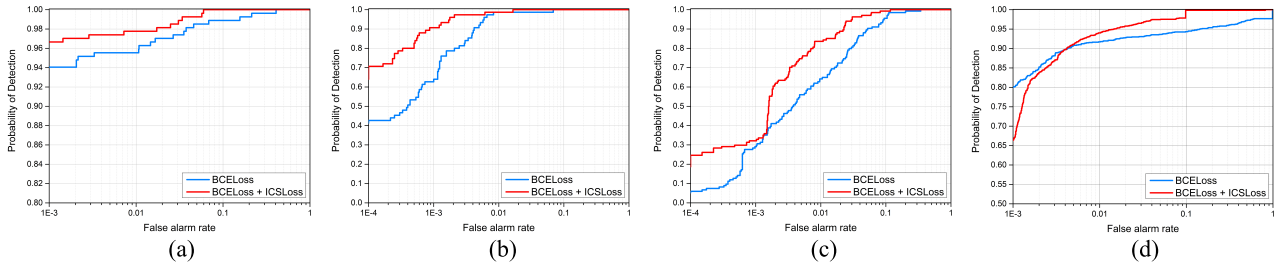

Fig. 11. Comparison of ROC curves with BCE loss and ICS loss for four data sets. (a) Muufl Gulfport. (b) Mosaic Avon. (c) San Diego. (d) Nuance Cri.

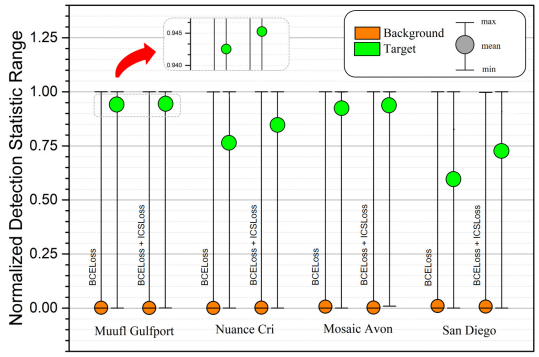

Fig. 12. Comparison of separability maps with BCE loss and ICS loss for four data sets.

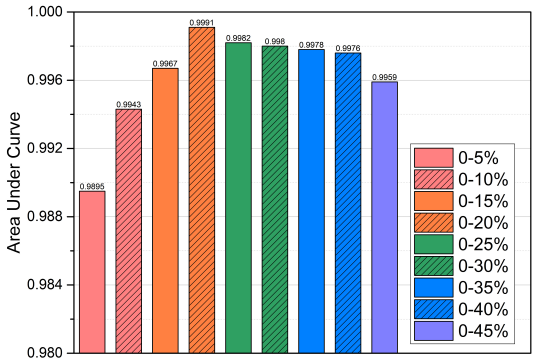

Fig. 13. AUC values of the TSCNTD with different abundance of background material for the Muufl Gulfport data set.

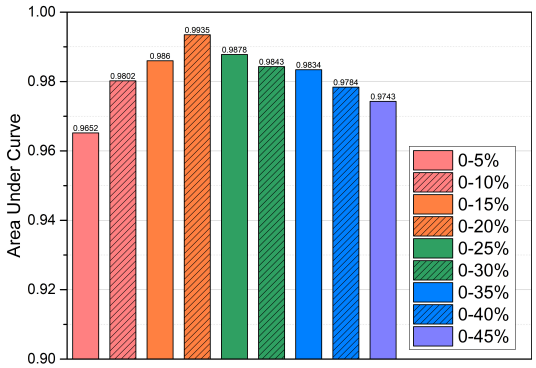

Fig. 14. AUC values of the TSCNTD with different abundance of background material for the San Diego data set.

separability maps with BCE loss and ICS loss for four HSIs is provided in Fig. 12.

According to Fig. 11, we can see that for the four HSI data sets, the ROC curves of the TSCNTD that contains BCE loss and ICS loss outperform those of the TSCNTD that only contains BCE loss due to the fact that the red curves are ahead of the blue curves. Although for the Nuance Cri data set the blue curve is ahead of the red one at the beginning, it falls a long way behind later. Therefore, the TSCNTD with BCE loss and ICS loss can achieve better performance. In Fig. 12, it can be seen that the TSCNTD can effectively suppress the background as the orange boxes are low for all data sets. For the target boxes, the range of the detector with BCE loss and ICS loss is always higher than the detector with only BCE loss, which indicates that the TSCNTD with BCE loss and ICS loss performs better in separability. Our proposed ICS loss helps reduce the distance between the target priori pixels and target pixels as well as increasing it between the target priori pixels and background pixels, which can enhance the separability between targets and backgrounds to improve the performance in target detection.

E. Analysis of Target Sample Generation

To estimate the influence of the abundance of background material that implanted to generate target samples, we used Muufl Gulfport and San Diego data sets and roughly set the abundance of background material implanted in the target priori pixels as [0%–5%, 0%–10%, 0%–15%, 0%–20%, 0%–25%, 0%–30%, 0%–35%, 0%–40%, 0%–45%] to test the detection performance of the TSCNTD. Figs. 13 and 14 present the AUC values of the TSCNTD for two data sets. The detection result maps of the TSCNTD for two data sets are provided in Figs. 15 and 16.

According to Figs. 13 and 14, it is obvious that the performance of the detector enhances as the range of background material abundance improves to a certain level. However, when the range of background material abundance is too large, bad detection performance is achieved. The reason is that the abundance of background material determines the number of subpixel targets that can be detected. When the range is small, pure target pixels and subpixel targets with few background materials can be detected. When the range is set too large, some background pixels that contain material that is similar to the target material may be regarded as subpixel targets and then highlight them, which causes

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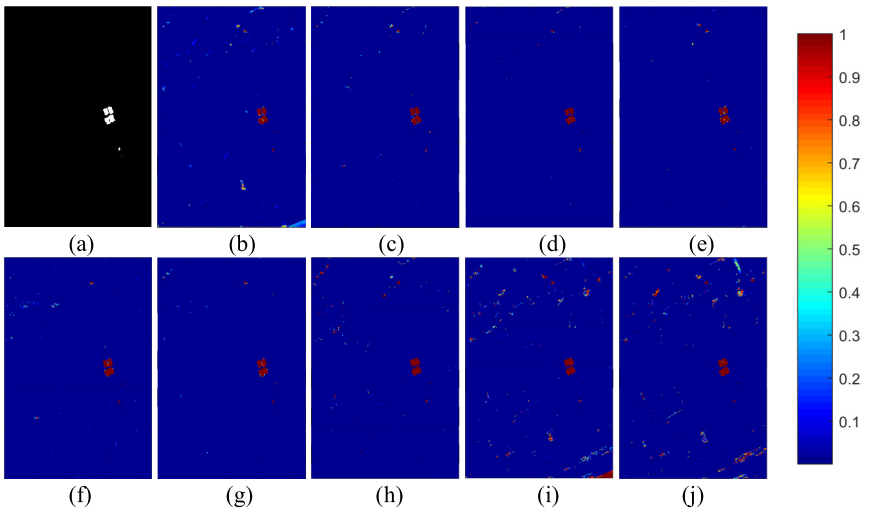

Fig. 15. Detection result maps of the TSCNTD with different background material abundance for the Muufl Gulfport data set. (a) Ground truth. (b) 0%–5%. (c) 0%–10%. (d) 0%–15%. (e) 0%–20%. (f) 0%–25%. (g) 0%–30%. (h) 0%–35%. (i) 0%–40%. (j) 0%–45%.

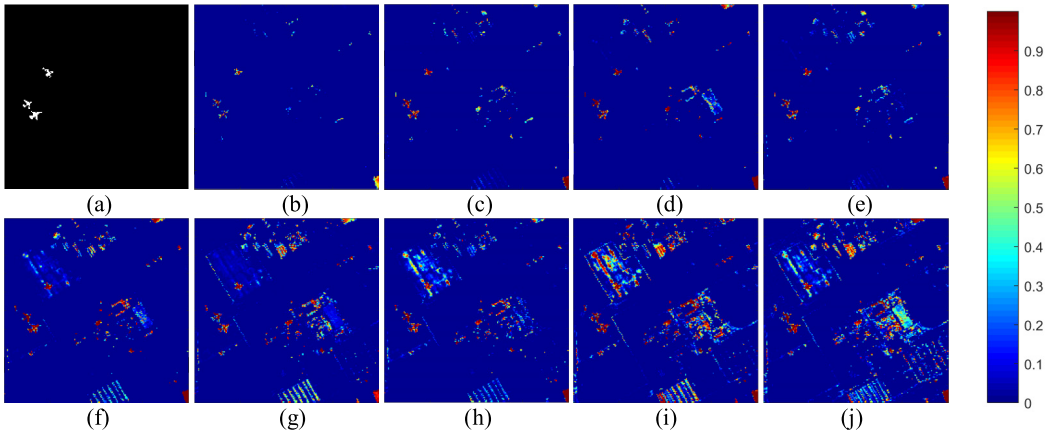

Fig. 16. Detection result maps of the TSCNTD with different background material abundance for the San Diego data set. (a) Ground truth. (b) 0%–5%. (c) 0%–10%. (d) 0%–15%. (e) 0%–20%. (f) 0%–25%. (g) 0%–30%. (h) 0%–35%. (i) 0%–40%. (j) 0%–45%.

more false alarms and leads to bad detection performance. The detection result maps of the TSCNTD with different background material abundances for two HSIs provided in Figs. 15 and 16 prove the case. When the range is small, i.e., 0%–5%, some of the subpixel targets cannot be detected. More subpixel targets can be detected as the range increases. When the range is too large, i.e., 0%–45%, some background pixels are detected as targets as they are mistakenly highlighted. Therefore, we set the abundance of background material as 0%–20% in this article, which can simultaneously detect most targets and control the false alarms.

F. Analysis of Time Cost

In this section, we analyze the time cost of the proposed method. The execution time of deep learning-based detectors is provided in Table II. From the perspective of the data size, a larger size of the image with more bands costs more time since the designed network needs to extract more features. The CNNTD costs more time than the HTD-Net and the proposed TSCNTD in both training and testing. The main reason is that the CNNTD needs more convolutional layers and utilizes a larger kernel size for feature extraction. For the HTD-Net, since the number of training samples is more than that in the TSCNTD, the run time of the HTD-Net is larger than that of the TSCNTD. For the proposed method TSCNTD, although

TABLE II
EXECUTION TIME (IN SECONDS) COMPARISON OF DEEP LEARNING-BASED DETECTORS FOR FOUR DATA SETS

Methods		Muufl Gulfport	Mosaic Avon	San Diego	Nuance Cri
CNNTD	Train	726.84 ±4.56	3363.67 ±20.96	1809.66 ±9.36	691.52 ±16.42
	Test	81.56 ±0.81	67.07 ±0.54	74.40 ±0.63	98.46 ±1.27
HTD-Net	Train	441.70 ±4.70	1039.91 ±18.16	581.25 ±7.51	437.40 ±2.91
	Test	6.22 ±0.35	174.51 ±1.07	64.66 ±0.57	9.87 ±0.61
TSCNTD	Train	63.77 ±1.79	133.99 ±1.94	56.65 ±0.81	62.92 ±1.78
	Test	1.36 ±0.05	5.34 ±0.18	6.87 ±0.05	1.76 ±0.25

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it is little bit time consuming in network training, it saves much more time compared with the CNNTD and the HTD-Net. Also, the testing time of the proposed TSCNTD is very short since several seconds are needed for testing.

V. Conclusion

Although many attempts have been made in HSI target detection, the detection performance of the present-day target detectors is still barely satisfactory. This article provides a deep network-based target detector the TSCNTD method for HSI target detection. The TSCNTD uses a hybrid sparse representation and classification-based pixel selection strategy to select the background pixels, which can ensure background samples are complete and pure. To solve the limited target samples, a new synthesis method is proposed to generate sufficient target samples using typical background pixels and target priori. The TSCNTD utilizes these obtained samples to construct the training samples. Then, a well-designed two-stream convolutional neural network of the TSCNTD is utilized to learn the discriminative ability. To separate the target pixels from backgrounds, we design an ICS loss in the proposed network, which can reduce the intraclass distance and increase the interclass distance. The ICS loss helps separate targets from backgrounds and enhances the final detection performance. Extensive experiments on four HSIs indicate that the TSCNTD achieves optimal detection performances and superior separability compared with many other advanced detectors. In the future, we will focus on how to use spatial information of HSI for target detection based on deep learning methods.

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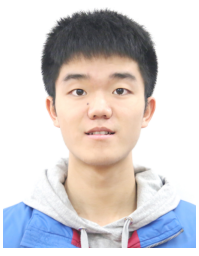

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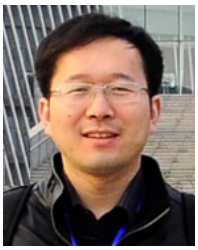

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